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STANDARDS

## AES standard for digital audio - Audio-embedded metadata - Part 4: Dolby E



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# AES standard for digital audio - Audio-embedded metadata - Part 4: Dolby E

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**Abstract**
AES41 provides for the carriage of audio metadata by embedding it in the audio samples themselves. This tightly associates the metadata with the audio, yet makes it fragile so that changes to the audio will invalidate the metadata. Several metadata sets have been defined, covering applications such as cascaded compression (bit rate reduction), and loudness control.

This part describes the format for the data to be transmitted with audio to signal downmix coefficients, loudness, and channel configuration metadata as used Dolby E.



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## Contents

## Foreword

This foreword is not part of AES41-4-2012 *AES standard for digital audio - Audio-embedded metadata - Part 4: Dolby E*

This document describes a set of data that may be conveyed according to the method described in Part 1 of this Standard.

As predicted in the foreword to AES41-2000, digital compression techniques now dominate the broadcast television environment. In addition to the problems foreseen relating to cascaded compression, new problems have arisen because of the use of loudness control and surround sound with those digital compression techniques.

Metadata within the compressed audio bit-stream is used to control loudness and the mixing down of multi-channel surround sound to two-channel stereo. These metadata are usually known by terms such as "dialnorm", "prog_ref_level", and "downmix coefficients".

Whilst this might seem unrelated to the original scope of AES41, dealing with bit allocations and scale factors, it is simply another form of data that can affect a later encoding of the audio: this time is it is more macroscopic than microscopic.

The metadata is lost when the bit-stream is uncompressed unless provision is made to transport it or store it somewhere. Existing methods rely on non-audio mechanisms to convey the metadata alongside the audio, for example a serial data link like RS-422 and serial digital video SMPTE 259M, or a "chunk" in an audio file (for metadata that does not change).

This revision extends AES41 to include data formats for carrying this loudness and downmix metadata with the uncompressed PCM using the same transport mechanism as before. The metadata can therefore be carried in the audio to which it relates.

The draft of this document was developed by a writing group whose primary author was Andrew Mason.

John Grant
Chair, working group SC-04-03
2012-03

## Note on normative language

In AES standards documents, sentences containing the word “shall” are requirements for compliance with the document. Sentences containing the verb “should” are strong suggestions (recommendations). Sentences giving permission use the verb “may”. Sentences expressing a possibility use the verb “can”.

# AES standard for digital audio - Audio-embedded metadata - Part 4: Dolby E

## 0 Introduction

AES41 provides for the carriage of audio metadata by embedding it in the audio samples themselves. This tightly associates the metadata with the audio, yet makes it fragile so that changes to the audio will invalidate the metadata. Several metadata sets have been defined, covering applications such as cascaded compression (bit rate reduction), and loudness control.

This part describes the format for the data to be transmitted with audio to signal downmix coefficients, loudness, and channel configuration metadata as used Dolby E.

### 0.1 Rationale for part 4 of this standard

Unwanted loudness variations in broadcast audio have historically been the source of many audience complaints. Broadcasters have adopted numerous techniques to address the problem – one of which involves indicating the long-term average loudness of the audio. Dolby E is a technology developed to convey multi-channel audio using the AES3 digital audio interface. Multi-channel infra-structure is now much more commonly available, but while the requirement for Dolby E is declining, the need for the metadata persists. Conveying the metadata as part of the audio that it describes offers significant advantages over mechanisms such as embedding in vertical ancillary data in a video stream.

## 1 Scope

This document describes a format for the data to be transmitted to convey a subset of the data in Dolby E (as described in SMPTE RDD6-2008). This Part assumes that the transmission mechanism according to Part 1 of this Standard is used.

## 2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

**AES41-1-2012** *AES standard for digital audio - Audio-embedded metadata - Part 1: General*, Audio Engineering Society, New York, NY., US.

**ISO/IEC Ed.3 1991,** *Information technology - ISO 7-bit coded character set for information interchange*, International Standards Organization, Geneva, Switzerland.

# 3 Definitions and abbreviations

## 3.1 Definitions

**3.1.1**
**mixdown**
The process by which surround sound audio is converted into (two-channel) stereo, usually by adding a fraction of the surround channels and the center channel into the front left and front right channels.

**3.1.2**
**Low frequency effects/enhancement channel**
**LFE**
A limited bandwidth channel for low frequency audio, typically in a multichannel system.

**3.1.3**
**dialnorm**
an indication of the average loudness of the audio signal (sometimes specifically its dialog component) that may be used to effect a normalisation to another target loudness

## 3.2 Symbols

The mathematical operators used in this standard are similar to those used in the C programming language. The bitwise operators are defined assuming 2's-complement representation of integers. Numbering and counting loops generally begin from zero.

### 3.2.1 Arithmetic operators

**3.2.1.1**
`+`
addition

**3.2.1.2**
`++`
increment

**3.2.1.3**
`--`
decrement

**3.2.1.4**
`*`
multiplication

**3.2.1.5**
`%`
modulus operator, defined only for positive numbers

### 3.2.2 Relational operators

**3.2.2.1**
`<`
less than

**3.2.2.2**

`==`
equal to

**3.2.2.3**
`!=`
not equal to

### 3.2.3 Bitwise operators

`&`
AND

### 3.2.4 Assignment

`=`
assignment operator

### 3.3 Abbreviations

The following mnemonics are defined to describe the different data types used in the coded bit stream.

**3.3.1**
**`bslbf`**
bit string, left bit first, where "left" is the order in which bit strings are written in this standard.

> NOTE Bit strings are written as a string of logic 1's and logic 0's within single quote marks, for example, '`1000 0001`'. Blanks within a bit string are for ease of reading and have no significance.

**3.3.2**
**`simsbf`**
signed integer, most significant bit first

**3.3.3**
**`uimsbf`**
unsigned integer, most significant bit first

# 4 Dolby E audio minimal metadata

## 4.1 AES41 metadata type

The presence of this data in the stream shall be indicated by **`type`**=$5_{10}$, as described in Part 1 of this standard.

## 4.2 Syntax

The syntax of **`data()`** is specified in the box below in accordance with Part 1.

| Syntax | No. of bits | Mnemonic |
|---|---|---|
| `data()` | | |
| `{` | | |
| `    if (type==Dolby E audio minimal metadata){` | | |
| `        description_text` | 8 | uimsbf |
| `        acmod` | 3 | bslbf |
| `        cmixlev` | 2 | bslbf |
| `        surmixlev` | 2 | bslbf |
| `        lfeon` | 1 | bslbf |
| `        dialnorm` | 5 | uimsbf |
| `}` | | |
| `    for(p=data_bits%8; p != 0; p--)` | | |
| `        padding_bit` | 2 | bslbf |
| `}` | | |

## 4.3 Semantics

The data are a minimal subset of the Dolby E audio metadata, where:

| | | |
|---|---|---|
| **`description_text`** | multi-character text description of the associated programme. | |
| | $00_{16}$ | (NUL), indicates that no text description information is present; |
| | $01_{16}$ | reserved for future standardization |
| | $02_{16}$ | (STX), indicates that the next character is the start of the text description; |
| | $03_{16}$ | (ETX), indicates that the previous character was the end of the text description; |
| | $04_{16}$ to $1F_{16}$ | reserved for future standardization |
| | $20_{16}$ to $7E_{16}$ | convey the corresponding ISO/IEC 646 (ASCII) character. |
| | $7F_{16}$ to $FF_{16}$ | reserved. |
| **`acmod`** | indicates which channels are in use: | |
| | 000 | reserved for future standardization |
| | 001 | center |
| | 010 | left, right |
| | 011 | left, center, right |
| | 100 | left, right, surround |
| | 101 | left, center, right, surround |
| | 110 | left, right, left surround, right surround |

| | | |
|---|---|---|
| | `111` | left, center, right, left surround, right surround |
| `cmixlev` | indicates the downmix level of the center channel with respect to the left and right channels: | |
| | `00` | -3 dB |
| | `01` | -4,5 dB |
| | `10` | -6 dB |
| | `11` | reserved |
| `surmixlev` | indicates the downmix level of the surround channels with respect to the left and right channels: | |
| | `00` | -3 dB |
| | `01` | -4,5 dB |
| | `10` | -6 dB |
| | `11` | reserved |
| `lfeon` | indicates that the low-frequency effects channel is on when '1', that it is off when '0' | |
| `dialnorm` | sign-inverted loudness value.<br>EXAMPLE: **`dialnorm`** = 23 indicates a loudness of -23 LKFS, | |
| `padding_bit` | indicates bits that are added to bring the total number of bits in the **`data`** to a multiple of 8 to match the length field. | |

## 5 Data alignment to audio samples

When the audio coder control data is the Dolby E audio minimal data set, and the audio has not been previously encoded, then the first bit of the audio coder control data shall be embedded in the first audio sample to which it applies. In this case, the repetition rate of the audio coder control data shall be the same as the frame rate of any associated video signal; if there is no associated video, the repetition rate shall be that of the predominant video frame rate in the environment,.

EXAMPLE: in many territories, including Europe, the video frame rate is 25 Hz. In other countries, including the US and Japan, the video frame rate is 30/1,001 Hz (nominally 29,97Hz).

When the audio coder control data has been derived from a Dolby E audio stream as it is decoded, the first bit of the audio coder control data shall be embedded in the audio sample that is output by the decoder exactly one video frame after the start of the Dolby E audio frame entering the decoder. This matches the one frame delay of the Dolby E decode process.

# Annex A Bibliography

**SMPTE RDD 6-2008** *Registered Disclosure Document: Television – Description and Guide to the Use of the Dolby E Audio Metadata Serial Bitstream*. Society of Motion Picture and Television Engineers, White Plains, NY., US.